

Cover Page

Standard: BACB30VT Rev: (AP) 26-Jan-2015

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BOEING PART STANDARD
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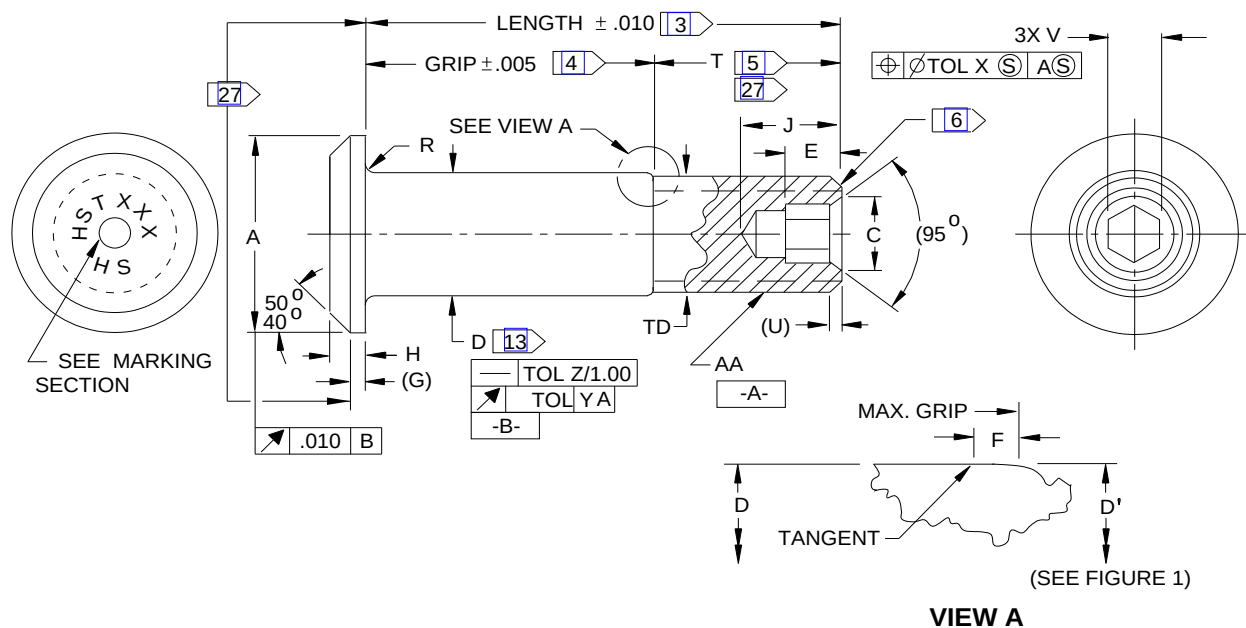
BACB30VT

Revision AP
26 January 2015
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**BOLT, PROTRUDING HEAD, HEX DRIVE, 95 KSI SHEAR, 6AL-4V TITANIUM, LIGHTWEIGHT
(NOMINAL AND OVERSIZE)**

TECHNICAL CHANGES IDENTIFIED BY REVISION BAR.

FOR STATUS OF INACTIVATION SEE APPLICABILITY SECTION.



DIMENSIONING AND TOLERANCING PER [ANSI Y14.5M-1982](#).

DIMENSIONS IN INCHES UNLESS OTHERWISE NOTED.

DIMENSIONS APPLY AFTER FINISH UNLESS OTHERWISE SPECIFIED.

TABLE I

BOEING STANDARD NUMBER BACB30VT [21]	AA THREAD SIZE [7]	$\varnothing$ A		$\varnothing$ C		$\varnothing$ D			
						NOMINAL SIZE			
						BEFORE FINISH	AFTER FINISH		
						ALUM COAT	ALL [2]	ALUM COAT, SOLID FILM	NONE OR PHOS FLUOR
		MAX	MIN	MAX	MIN	MIN	MAX	MIN	MIN
5	.1640-32	.262	.242	[1]	[1]	.1621	.1635	.1625	.1630
5H	.1640-32	.262	.242	[1]	[1]	.1621	.1635	.1625	.1630
6	.1900-32	.315	.295	.119	.104	.1881	.1895	.1885	.1890
8	.2500-28	.412	.387	.142	.122	.2481	.2495	.2485	.2490
10	.3125-24	.505	.475	.180	.160	.3106	.3120	.3110	.3115
12	.3750-24	.600	.565	.217	.197	.3731	.3745	.3735	.3740
14	.4375-20	.676	.641	.253	.233	.4356	.4370	.4360	.4365

TABLE I (Continued)

BOEING STANDARD NUMBER BACB30VT 21	Ø D							
	.0156 OVERSIZE 16				.0312 OVERSIZE 16			
	BEFORE FINISH	AFTER FINISH			BEFORE FINISH	AFTER FINISH		
	ALUM COAT	ALL 2	ALUM COAT, SOLID FILM	NONE OR PHOS FLUOR	ALUM COAT	ALL 2	ALUM COAT, SOLID FILM	NONE OR PHOS FLUOR
	MIN	MAX	MIN	MIN	MIN	MAX	MIN	MIN
5	17	17	17	17	17	17	17	17
5H	17	17	17	17	17	17	17	17
6	.2012	.2026	.2016	.2021	.2168	.2182	.2172	.2177
8	.2637	.2651	.2641	.2646	.2793	.2807	.2797	.2802
10	.3262	.3276	.3266	.3271	.3418	.3432	.3422	.3427
12	.3887	.3901	.3891	.3896	.4043	.4057	.4047	.4052
14	.4512	.4526	.4516	.4521	.4668	.4682	.4672	.4677

BOEING STANDARD NUMBER BACB30VT 21	Ø D' MIN			E MIN	F MAX	G REF	H		J MAX 15
	NOMINAL	.0156 OVERSIZE 16	.0312 OVERSIZE 16				MAX	MIN	
5	.1601	17	17	.115	.013	.020	.047	.037	.178
5H	.1601	17	17	.080	.013	.020	.047	.037	.140
6	.1857	.1990	.2044	.080	.016	.025	.055	.045	.140
8	.2451	.2609	.2763	.090	.021	.030	.069	.059	.200
10	.3068	.3226	.3380	.110	.026	.035	.078	.068	.240
12	.3693	.3851	.4005	.140	.030	.040	.088	.078	.275
14	.4318	.4476	.4632	.210	.035	.045	.105	.093	.320

BOEING STANDARD NUMBER BACB30VT 21	JT RAD (SEE Figure 1) 12		L MAX (SEE Figure 2) 19			R RAD		T 5		Ø TD 7		U REF
	MAX	MIN	NOMINAL	OVERSIZE		MAX	MIN	NOMINAL	OVERSIZE 16	MAX	MIN	
				.0156	.0312							
5	---	---	---	---	---	.025	.015	.280	17	.1595	.1570	.031
5H	.080	.070	.026	---	---	.025	.015	.280	17	.1595	.1570	.031
6	.100	.070	.026	.031	.036	.025	.015	.290	.300	.1840	.1810	.031
8	.140	.130	.027	.032	.037	.025	.015	.320	.330	.2440	.2410	.031
10	.170	.160	.029	.034	.039	.030	.020	.380	.390	.3060	.3020	.047
12	.230	.160	.029	.034	.039	.030	.020	.420	.430	.3680	.3640	.047
14	.300	.290	---	---	---	.030	.020	.485	.495	.4310	.4260	.047

TABLE I (Continued)

BOEING STANDARD NUMBER BACB30VT [21]	V HEX		TOLERANCE		
	MAX	MIN	X	Y	Z
5	.0645	.0635	.010	.0045	.0040
5H	.0806	.0791	.010	.0045	.0040
6	.0806	.0791	.010	.0045	.0040
8	.0967	.0947	.010	.0045	.0030
10	.1295	.1270	.012	.0045	.0030
12	.1617	.1582	.014	.0060	.0025
14	.1930	.1895	.017	.0060	.0025

NOTES

- [1] ON SIZE 5H ONLY, COUNTERSINK TO A DEPTH TO BREAK CORNERS OF THE DRIVE HEX.
- [2] MAXIMUM "D" DIAMETER MAY BE EXCEEDED BY .0002 FOR SOLID FILM LUBRICATED PARTS.
- [3] THE TOLERANCE SHALL BE APPLIED TO A NOMINAL LENGTH DIMENSION DETERMINED BY ADDING THE NOMINAL GRIP LENGTH (GRIP LENGTH NUMBER TIMES .0625) AND "T" FROM [Table I](#). [4] [5]
- [4] TO DETERMINE THE NOMINAL GRIP LENGTH, MULTIPLY THE GRIP LENGTH NUMBER BY .0625. THE GRIP LENGTH IS MEASURED FROM THE UNDERSIDE OF THE HEAD TO THE D' DIAMETER.
- TO DETERMINE THE GRIP LENGTH NUMBER, DIVIDE THE TOTAL THICKNESS OF PARTS BEING JOINED BY .0625. ROUND OFF DECIMALS TO NEXT LARGER WHOLE NUMBER.
[3] [10]
- [5] THE "T" DIMENSION IS NOMINAL AND DOES NOT HAVE AN APPLIED TOLERANCE. DIMENSION "T" IS NOT TO BE INSPECTED, BUT IS USED IN CALCULATION OF NOMINAL BOLT LENGTH. [3]
- [6] END SHALL BE FLAT AND CHAMFERED. CHAMFER "U" PLUS INCOMPLETE THREAD NOT TO EXCEED TWO THREAD PITCHES.
- [7] THREAD CLASS UNJF-3A EXCEPT UNJC-3A FOR SIZE 5 OR 5H (.1640-32) PER [AS8879](#). MAJOR DIAMETER SHALL CONFORM TO "TD".
- [8] SEE DESIGN CAUTION NOTE UNDER USAGE AND APPLICATION.
- [9] THE INTERSECTION BETWEEN THE SHANK AND THREAD OF HI-LITE TYPE BOLTS (HST PART NUMBERS IN [Table II](#)) SHALL BE COLD WORKED SUBSEQUENT TO HEAT TREATMENT. THIS PROCESS IS NOT REQUIRED FOR VERI-LITE TYPE BOLTS (VL PART NUMBERS IN [Table II](#)), AERO LITE TYPE BOLTS (AL PART NUMBERS IN [Table II](#)), OR ULTRA-LITE TYPE BOLTS (WC PART NUMBERS IN [Table II](#)).

- 10 GRIP LENGTH, TRANSITION RADIUS AND LEAD AND RUNOUT THREADS SHALL BE INSPECTED USING COMPARATOR AND OVERLAY CHARTS PER [Figure 1](#). 14
- 11 THREADS ADJACENT TO CHAMFER AND ADJACENT TO GRIP MAY BE INCOMPLETE. INCOMPLETE THREADS MAY BE OUT OF TOLERANCE AT MAJOR, PITCH, OR MINOR DIAMETER, BUT LENGTH OF INCOMPLETE THREADS AT LEAD SHALL NOT EXCEED 2 PITCHES AND LENGTH AT GRIP SHALL NOT EXCEED 1.5 PITCHES FOR NOMINAL SIZE BOLTS AND 2 PITCHES FOR OVERSIZE BOLTS. ADDITIONALLY, THE ROOT OF THE RUNOUT THREADS SHALL HAVE RADII OF .15011 PITCH MINIMUM, BLENDING SMOOTHLY TO THE THREAD FLANKS.
- 12 "JT" TRANSITION RADIUS MUST FAIR INTO SHANK DIAMETER WITH NO VISIBLE LINE WITHIN "F" DISTANCE, AND EXTEND BELOW THE "TD" MAXIMUM DIAMETER PER [Figure 1](#) FOR NOMINAL SIZE BOLTS. THE TRANSITION RADIUS FOR OVERSIZE BOLTS SHALL EXTEND BELOW THE SHANK DIAMETER THE SAME MINIMUM RADIAL DISTANCE AS THE NOMINAL SIZE BOLTS.
- 13 RUNOUT OF THE "D" DIAMETER TO THE THREAD PITCH DIAMETER SHALL BE WITHIN "Y" FIM WHEN HELD ON THE PITCH DIAMETER OF THE COMPLETE THREADS NEAREST THE SHANK AND CHECKED WITHIN ONE DIAMETER OF THE THREAD RUNOUT.
- 14 ALL LOTS PRODUCED AFTER JUNE 30, 1995 SHALL HAVE THIS MARKING. PARTS IN THE SYSTEM WITH THE OLD DEUTSCH SYMBOL OF DA OR DF ARE ACCEPTABLE.
- 15 MAXIMUM "J" DIMENSION OF .178 IS ACCEPTABLE FOR SIZE 5H AND 6 BOLTS PRODUCED BEFORE MAY 1, 1995 PROVIDED THEY MEET ALL MECHANICAL STRENGTH REQUIREMENTS OF [BPS-F-67](#).
- 16 RESTRICTED USAGE: FOR REPAIR ONLY.
- 17 NO SIZE 5 AND 5H OVERSIZE. SEE USAGE AND APPLICATION SECTION FOR SIZE 5 AND 5H REPAIR INFORMATION.
- 18 HUCK LAKEWOOD (FORMERLY DEUTSCH FASTENER CORP) IS NO LONGER IN OPERATION. DIAMETERS 5 THRU 14, MANUFACTURED AT THAT FACILITY (CAGE CODE 97928) UNDER EITHER NAME PRIOR TO OCTOBER 1, 1999 MAY BE PROCURED AND USED UNTIL STOCKS ARE DEPLETED.
- 19 PARTS RELEASED FOR PRODUCTION (AT THE SUPPLIER'S FACILITY) AFTER JUNE 1, 1999 SHALL MEET THE FOLLOWING REQUIREMENTS FOR TRANSITION RUN OUT LIMITS (SIZES 5H THROUGH 12 ONLY):
- FINGER TIGHTEN A FACED OFF GO RING GAGE ON THE THREADS AND MEASURE DIMENSION "M". DIMENSION "M" MUST BE LESS THAN NOMINAL GRIP LENGTH + "L" MAXIMUM (SEE [Figure 2](#) AND [Table I](#)). ACCEPTANCE SAMPLING PER "ATTRIBUTE SAMPLING FOR FINISH INSPECTION AND COMPARATOR MEASUREMENTS" OF [BPS-F-67](#).
- 20 THE SURFACE OF THE RING GAGE SHALL BE GROUND SUCH THAT THE LEAD-IN CHAMFER IS REMOVED (SHARP THREAD START), CONFORMING TO ST1416-13 DRAWING.
- 21 SEE CODING UNDER USAGE AND APPLICATION FOR COMPLETE BOEING PART NUMBER.

- [22] PARTS RELEASED FOR PRODUCTION (AT THE SUPPLIER'S FACILITY) AFTER JUNE 1, 1999 SHALL HAVE THESE PART NUMBERS. PARTS PREVIOUSLY MADE AS VL310-5 AND VL10 MAY BE USED UNTIL STOCK IS DEPLETED.
- [23] ALCOA FASTENING SYSTEMS (TEMPLE) (FORMERLY FAIRCHILD - TEMPLE) IS NO LONGER IN OPERATION. PARTS MANUFACTURED AT THAT FACILITY (CAGE CODE 06950) UNDER EITHER NAME PRIOR TO SEPTEMBER 1, 2006 MAY BE PROCURED AND USED UNTIL STOCKS ARE DEPLETED.
- [24] FAIRCHILD FASTENERS (UNRUH) IS NOW ALCOA FASTENING SYSTEMS - UNRUH. THIS WAS A NAME CHANGE ONLY. PARTS MANUFACTURED UNDER THE FAIRCHILD NAME MAY BE PROCURED AND USED UNTIL STOCKS ARE DEPLETED.
- [25] HUCK INTERNATIONAL INC (CARSON) IS NOW ALCOA FASTENING SYSTEMS - CARSON. THIS WAS A NAME CHANGE ONLY. PARTS MANUFACTURED UNDER THE HUCK NAME MAY BE PROCURED AND USED UNTIL STOCKS ARE DEPLETED.
- [26] SPS TECHNOLOGIES, KEARNS IS NO LONGER IN OPERATION. PARTS MANUFACTURED AT THAT FACILITY (CAGE CODE 0PTK6) PRIOR TO JUNE 30, 2002 MAY BE PROCURED AND USED UNTIL STOCKS ARE DEPLETED.
- [27] FOR "B" FINISH CODE, BMS10-85 COATING SHALL BE APPLIED TO THE TOP OF THE HEAD AND THREADS ONLY. INCOMPLETE COVERAGE IS ALLOWED ON THE RUNOUT THREADS STARTING FROM THE GRIP SECTION OF THE SHANK, EXTENDING INTO THE THREADS A MAXIMUM OF .030. NO OVERSPRAY IS ALLOWED ON THE CYLINDRICAL PORTION OF THE SHANK. OVERSPRAY OF .005 ON THE HEAD BEARING SURFACE IS ACCEPTABLE.

PROCUREMENT SPECIFICATION

[BPS-F-67](#), EXCEPT AS NOTED.

FOR "F" CODE PARTS THERE SHALL BE NO LAPS ALLOWED ON THE NON-BEARING SURFACES OF THE HEAD.

MATERIAL

6Al-4V TITANIUM ALLOY PER [AMS 4928](#) OR [AMS 4967](#).

MINIMUM SHEAR STRENGTH: 95 KSI.

FINISH

ALUMINUM PIGMENTED COATING PER [BMS10-85](#), TYPE I, CLASS A

PHOSPHATE-FLUORIDE TREAT PER [BAC5861](#). [8]

SURFACE TEXTURE

PER [ASME B46.1](#). HEAD TO SHANK FILLET, THREAD ROOT, THREAD FLANKS AND SHANK, 32 MICROINCHES Ra; BEARING SURFACE OF HEAD, 63 MICROINCHES Ra; OTHER SURFACES, 125 MICROINCHES Ra.

LUBRICATION

CETYL ALCOHOL IN ACCORDANCE WITH [AS87132](#), TYPE I OR TYPE III, GRADE OPTIONAL.

SOLID FILM LUBE PER [AS5272](#), TYPE I.

MARKING

MARK BOLT HEAD WITH MANUFACTURER'S BASIC NUMBER AND SYMBOL/ INSIGNIA PER [Table II](#), [MIL-HDBK-57](#), OR REGISTERED WITH THE U.S. PATENT AND TRADEMARK OFFICE (PTO) OF THE U.S. DEPARTMENT OF COMMERCE. MARKING WITH THE FIRST DASH NUMBER IS OPTIONAL. MARKING INDENTED .010 MAXIMUM, ARRANGEMENT OPTIONAL. CODE LETTER "F" PARTS SHALL ALSO BE MARKED WITH A RAISED OR DEPRESSED CIRCLE (.005 MAXIMUM) LOCATED APPROXIMATELY AS SHOWN IN THE GRAPHIC.

IDENTIFY WITH ORANGE DYE OR PAINT ON THE THREAD END EXCEPT FOR "B" CODE ONLY, APPLY WHITE DYE OR PAINT ON THE THREAD END. COVERAGE OF DYE OR PAINT MAY INCLUDE THE CHAMFER PLUS ONE PITCH. APPLY BEFORE LUBRICATION WITH CETYL ALCOHOL OR AFTER SOLID FILM LUBE.

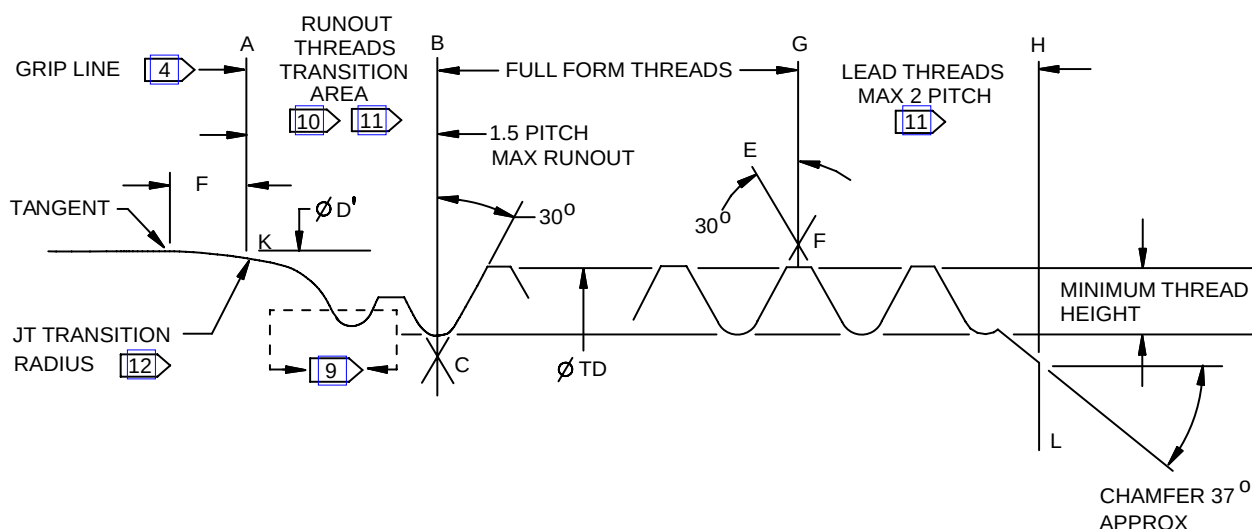
INSPECTION REQUIREMENTS

FIGURE 1 - DETERMINATION OF INCOMPLETE THREADS BY COMPARATOR

CHART REQUIREMENTS ([Figure 1](#))

1. FOR RUNOUT THREADS: HORIZONTAL PARALLEL LINES TO DEFINE MINIMUM THREAD HEIGHT, WHICH IS ONE-HALF THE DIFFERENCE BETWEEN MINIMUM MAJOR DIAMETER AND MAXIMUM MINOR DIAMETER; VERTICAL LINE "AK" LOCATED TO LEFT OF VERTICAL LINE "BC"; VERTICAL LINE "BC" INTERSECTING 30 DEGREE FLANK ANGLE LINE "CD" AT POINT "C".
2. FOR LEAD THREADS: HORIZONTAL LINES AS DESCRIBED ABOVE. VERTICAL LINE "GF" INTERSECTING 30 DEGREES FLANK ANGLE LINE "EF" AT POINT "F". VERTICAL LINE "HL" LOCATED 2 PITCH DISTANCE TO RIGHT OF "GF".

PROCEDURES ([Figure 1](#))

1. FOR RUNOUT THREADS: ROTATE FASTENER TO FIND FIRST COMPLETE THREAD NEAREST TO GRIP. AT THIS POINT, THREAD HEIGHT EQUALS MINIMUM HEIGHT AS INDICATED ON CHART. THEN MOVE SHADOW LONGITUDINALLY UNTIL LEFT FLANK OF FIRST COMPLETE THREAD PROFILE COINCIDES WITH LINE "CD". RUNOUT THREADS SHALL THEN FALL BETWEEN LINE "AK" AND "BC" TO BE ACCEPTABLE.

2. FOR LEAD THREADS: ROTATE FASTENER TO FIND FIRST COMPLETE THREAD NEAREST TO END. AT THIS POINT COMPLETE THREAD HEIGHT EQUALS MINIMUM HEIGHT AS INDICATED ON CHART. THEN MOVE SHADOW LONGITUDINALLY UNTIL RIGHT FLANK OF FIRST COMPLETE THREAD PROFILE COINCIDES WITH LINE "EF". LEAD THREADS SHALL THEN FALL BETWEEN LINES "GF" AND "HL" TO BE ACCEPTABLE.

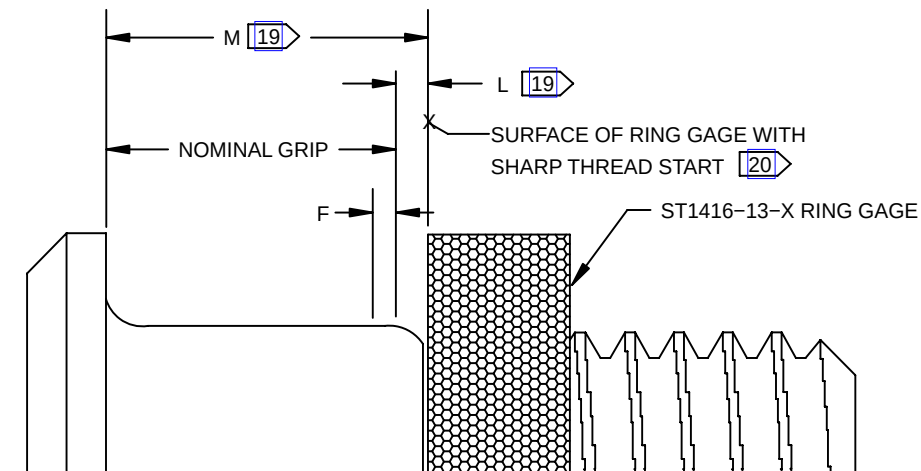


FIGURE 2 - TRANSITION RUNOUT LIMITS

PROCUREMENT

AIR INDUSTRIES CORP (CAGE CODE 06725)

ALCOA FASTENING SYSTEMS - CARSON (CAGE CODE 17446) [25]

ALCOA FASTENING SYSTEMS - INDUSTRY (TEMPLE) (CAGE CODE 06950) [23]

ALCOA FASTENING SYSTEMS - INDUSTRY (UNRUH) (CAGE CODE 1RC86) [24]

LISI AEROSPACE - HI-SHEAR PRODUCTS (CAGE CODE 73197)

HUCK INTERNATIONAL INC, LAKEWOOD OPERATIONS (CAGE CODE 97928) [18]

SPS TECHNOLOGIES, AEROSPACE PRODUCTS DIV (JENKINTOWN) (CAGE CODE 56878)

SPS TECHNOLOGIES, AEROSPACE PRODUCTS DIV (KEARNS) (CAGE CODE 0PTK6) [26]

WEST COAST AEROSPACE, INC. (CAGE CODE 60516)

THE MANUFACTURERS LISTED IN [BPS-F-67SUP](#) AND THEIR AUTHORIZED DISTRIBUTORS ARE THE ONLY APPROVED SOURCES FOR THE ABOVE QUALIFIED PRODUCTS. SEE [BPS-F-67SUP](#) FOR PLANT ADDRESSES. NO CHANGES IN PRODUCT DESIGN, BASIC METHODS OF MANUFACTURE, PLANT SITE OR QUALITY LEVEL SHALL BE MADE WITHOUT PRIOR NOTIFICATION AND PRIOR APPROVAL IN WRITING FROM THE BOEING COMPANY. MANUFACTURERS OF COMPETITIVE PRODUCTS MAY APPLY TO A SUPPLIER MANAGEMENT DEPARTMENT OF THE BOEING COMPANY FOR QUALIFICATION. IF A MANUFACTURER IS SHOWN ON THIS STANDARD, BUT NOT LISTED IN THE SUPPLEMENT, CONTACT THE DIVISIONAL ENGINEERING STANDARDS FOCAL POINT OR ENGINEERING STANDARDS FOR VERIFICATION.

THIS IS A MANUFACTURER-DESIGNED PRODUCT. BOEING MAKES NO REPRESENTATION WHATEVER REGARDING PATENT OR OTHER RIGHTS AFFECTING THE PRODUCT. THE LISTING OF ANY SUPPLIER DOES NOT IMPLY ANY DETERMINATION BY THE BOEING COMPANY OR BY ANY OTHER LISTED MANUFACTURER AS TO THE RIGHTS OF SUCH MANUFACTURER.

HI-SHEAR CORP STATES THAT PATENT NUMBER 4,326,825 IS PERTINENT TO ITS PRODUCT.

USAGE AND APPLICATION INFORMATION

THESE BOLTS AND MATING COLLARS, WITH COUNTERBORE, ARE DESIGNED SO THAT NOMINAL GRIP DIMENSION EQUALS MAXIMUM MATERIAL THICKNESS (OR MAXIMUM GRIP RANGE). WHEN MATERIAL THICKNESS EQUALS MINIMUM GRIP, .062 NOMINAL OF UNTHREADED SHANK ENTERS COUNTERBORE OF NUT. THIS DESIGN MAKES THE GRIP RANGE OF THIS FASTENER EQUAL TO THE GRIP RANGE OF SHEAR LOCKBOLTS OR HI-SHEAR RIVETS OF THE SAME DASH NUMBER. "F" CODE PARTS ARE INTENDED FOR HIGH VIBRATION APPLICATIONS SUCH AS HELICOPTER COMPOSITE STRUCTURE.

INSTALL PER [BAC5004-2](#), [BAC5063](#), OR [BAC5063-2](#), AS APPLICABLE.

CAUTION

TEST DATA INDICATES THAT PHOSPHATE FLUORIDE COATED FASTENERS INSTALLED IN CARBON-EPOXY STRUCTURE CAN BE POTENTIALLY HARMFUL TO THE STRUCTURE IF EXPOSED TO A SALT ENVIRONMENT.

OVERSIZE BOLTS

BH ONLY

CODES "X" AND "Y" ARE USED AS REPLACEMENTS FOR STANDARD SIZE BACB30VT BOLTS HAVING THE SAME FINISH WHEN THE HOLE FOR THE STANDARD SIZED BOLT IS OVERSIZE OR MUST BE MADE OVERSIZE TO CORRECT DAMAGE OR MISALIGNMENT.

TO REPLACE BACB30VT5() () OR BACB30VT5H() () BOLTS, USE BACB30VT6() () FOR FIRST OVERSIZE AND BACB30VT6() ()X FOR SECOND OVERSIZE.

BCA ONLY

SEE [BAC5004-2](#) FOR OVERSIZE PART INFORMATION.

BDS ONLY (EXCEPTIONS: BH)

OVERSIZE BOLTS [BACB30MY](#)() () X (.0156 OVERSIZE) AND [BACB30MY](#)() () Y (.0312 OVERSIZE) ARE TO BE USED AS REPLACEMENTS FOR STANDARD SIZE BACB30VT BOLTS HAVING THE SAME FINISH WHEN THE HOLE FOR THE STANDARD SIZED BOLT IS OVERSIZE OR MUST BE MADE OVERSIZE TO CORRECT DAMAGE OR MISALIGNMENT. CARE SHOULD BE TAKEN TO ASSURE THE PROPER COLLAR IS SELECTED PER [BAC5004-2](#).

TO REPLACE BACB30VT5() () OR BACB30VT5H() () BOLTS, USE BACB30VT6() () FOR SECOND OVERSIZE AND BACB30NX6() () X FOR THIRD OVERSIZE. THERE IS NO FIRST OVERSIZE.

CODING

FIRST DASH NUMBER DESIGNATES NOMINAL THREAD SIZE IN .03125 INCREMENTS.

SUFFIX "H" TO FIRST DASH NUMBER DESIGNATES ENLARGED V HEX (FOR SIZE 5 ONLY).

CODE FOLLOWING FIRST DASH NUMBER DESIGNATES FINISH AND LUBRICATION AS FOLLOWS:

"-" DESIGNATES NO FINISH AND CETYL ALCOHOL LUBE.

"B" DESIGNATES ALUMINUM PIGMENTED COATING APPLIED TO THE HEAD AND THREADS ONLY, AND CETYL ALCOHOL LUBRICATION. [127](#)

"K" DESIGNATES ALUMINUM PIGMENTED COATING AND CETYL ALCOHOL LUBE.

"R" DESIGNATES PHOSPHATE-FLUORIDE TREATMENT AND CETYL ALCOHOL LUBE.

"S" DESIGNATES PHOSPHATE-FLUORIDE TREATMENT AND SOLID FILM LUBE.

SECOND DASH NUMBER FOLLOWING FINISH AND LUBRICATION CODE DESIGNATES NOMINAL GRIP LENGTH IN .0625 INCREMENTS. [4](#)

"F" CODE FOLLOWING SECOND DASH NUMBER DESIGNATES A LAP-FREE HEAD.

LETTER "X" AT THE END OF THE PART NUMBER DESIGNATES .0156 OVERSIZE. [16](#) [17](#)

LETTER "Y" AT THE END OF THE PART NUMBER DESIGNATES .0312 OVERSIZE. [16](#) [17](#)

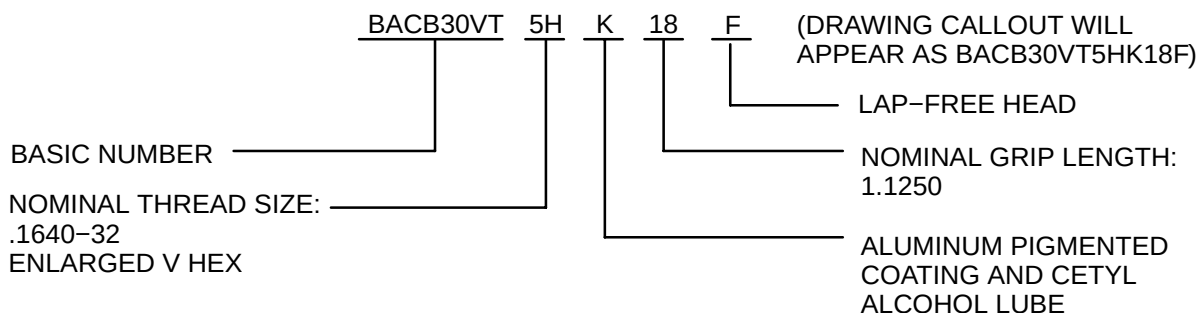
EXAMPLE OF PART NUMBERFASTENER CODESEE [BACD2074](#) FOR FASTENER CODES.

TABLE II

MANUFACTURER	MANUFACTURER'S BASIC NUMBER			SYMBOL/INSIGNIA
	NOMINAL SIZE	.0156 OVERSIZE	.0312 OVERSIZE	
AIR INDUSTRIES	HST10 EXCEPT HST108 FOR SIZE 5H ONLY	HST110	HST410	
HI-SHEAR				HS
SPS				SPS OR S
ALCOA FASTENING SYSTEMS	VL10 EXCEPT VL310-5 FOR SIZE 5H ONLY	VL110	VL410	O OR VS
HUCK	AL10 EXCEPT AL108 FOR SIZE 5H ONLY	AL110	AL410	
WEST COAST AEROSPACE	WC10 EXCEPT WC108 FOR SIZE 5H ONLY	WC10X	WC10Y	WC
MANUFACTURER'S BASIC NUMBER AND SYMBOL/INSIGNIA LISTED ARE FOR HEAD MARKING AND ARE NOT TO BE USED FOR DRAWING CALLOUT.				

D-590 NOTESSEE [D-590-PREFACE](#) FOR INACTIVATION DEFINITIONS. SEE [00000-D-590-INFO](#) FOR LINKS TO SUPERSESSION CLASS DEFINITIONS AND FAQ.**INACTIVATION APPLICABILITY**

BCA

- BACB30VT()R() IS INACTIVE FOR NEW DESIGN.

BACB30VT5(ALL CODES)() ARE INACTIVE FOR DESIGN AND PROCUREMENT.

BACB30VT5H(ALL CODES)() ARE CLASS II SUPERSESSIONS, RESPECTIVELY.

BCA	- BACB30VT14(ALL CODES)() ARE INACTIVE FOR DESIGN AND PROCUREMENT. NO SUPERSEDING PARTS. BACB30VT()-() IS INACTIVE FOR DESIGN AND PROCUREMENT. NO SUPERSEDING PARTS. SEE BACB30VT()K(). BACB30VT()(ALL CODES)()F ARE INACTIVE FOR DESIGN AND PROCUREMENT. NO SUPERSEDING PARTS. BACB30VT()(ALL CODES)()(X, Y) ARE INACTIVE FOR DESIGN AND PROCUREMENT. NO SUPERSEDING PARTS. BACB30VT()B() IS INACTIVE FOR DESIGN AND PROCUREMENT. NO SUPERSEDING PARTS.
BDS (EXCEPTIONS: BH)	- BACB30VT()(ALL CODES)()F ARE INACTIVE FOR DESIGN AND PROCUREMENT. NO SUPERSEDING PARTS. BACB30VT()(ALL CODES)()(X, Y) ARE INACTIVE FOR DESIGN AND PROCUREMENT. NO SUPERSEDING PARTS. BACB30VT5(ALL CODES)()(NO CODE, F) ARE INACTIVE FOR DESIGN AND PROCUREMENT. BACB30VT5H(ALL CODES)()(NO CODE, F) ARE CLASS IIA SUPERSESSIONS, RESPECTIVELY.
BH	- BACB30VT()R()(NO CODE, F) ARE INACTIVE FOR DESIGN AND PROCUREMENT. BACB30VT()-()(NO CODE, F) ARE CLASS II SUPERSESSIONS, RESPECTIVELY. BACB30VT5(ALL CODES)()(NO CODE, F) ARE INACTIVE FOR DESIGN AND PROCUREMENT. BACB30VT5H(ALL CODES)()(NO CODE, F) ARE CLASS II SUPERSESSIONS, RESPECTIVELY.